



Mohamed Nisfan

Science and Technology undergraduate specializing in Mechatronics with hands-on experience in CAD design, Machine Learning, Deep Learning, Embedded systems, industrial automation, and web development. Skilled in combining mechanical, electronic, computational, and frontend systems for real-time applications. Experienced in using Arduino, SolidWorks, AutoCAD, and modern web technologies for design, prototyping, and application development. Passionate about developing intelligent, data-driven systems that improve machine performance, reliability, and user interaction.

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EDUCATION

- Bsc. Hons. in Science and Technology (Mechatronics specialization)

Uva Wellassa University - Badulla
2021 SEP / 2025 AUG

- Diploma in Management

Uva Wellassa University - Badulla
2022 MAY / 2023 MAY

- Certificate in Human Resource Management

Chartered Institute of Personnel Management - Sri Lanka
2020 JUN / 2020 DEC

PROJECTS

- Real-Time Machine Health Monitoring Using Deep Boltzmann Machine

A model to monitor system health using voltage, current, and heat sensors. Utilized a Deep Boltzmann Machine for analyzing collected data.

Embedded | Python | Arduino IDE | Deep Learning (DBM) | Sensors | Firebase

- SymptoGuide : Human Health Monitoring Website (Ongoing)

A healthcare website to assist patients in understanding disease symptoms and improvements. Multilingual support has been implemented using ReactJS and i18n. A machine learning model has to be integrated for prediction purpose

ReactJS | i18n | JavaScript | Machine Learning | Flask & Express

- Machinery Identification using CNN (OpenCV)

Developed a machine learning-based image classification model to identify workshop machinery using CNNs. Trained with augmented image datasets and deployed for real-time predictions using OpenCV and Keras.

Python | TensorFlow | Keras | OpenCV | CNN | Image Classification | MachineLearning

- Mobile Controlled Home Automation System

A simple model was developed with the help of ESP-32, and Relays etc. Using Python a simple GUI was made to control the module

Embedded | IoT Integration | Microcontroller Programming

- Online Printer Management System using Python

Developed an intuitive web-based printer management system using Django, Python, HTML, and CSS, featuring secure authentication, file uploads, customizable printing options, dynamic cost calculation, and real-time print job monitoring.

Python | HTML | CSS | Javascript

- Fuel Distribution Management System using C

Developed an innovative C-based application to streamline fuel distribution during Sri Lanka's COVID-19 fuel crisis, featuring real-time updates, efficient resource allocation, and crisis management tools.

C Language

- Sign Language Detection using Python & Scikit-learn

Developed a machine learning-based application to recognize and interpret sign language gestures using Python and Scikit-learn, enabling real-time gesture classification for improved communication accessibility.

Python | OpenCV | MediaPipe | Scikit-learn | Numpy

- Library Management CRUD App using Next.JS

Developed a full-stack Library Management System using Next.js to perform efficient CRUD operations for managing book records, with a focus on modular UI design and dynamic data handling.

Next.js | React | Tailwind CSS | Node.js | MongoDB | Mongoose

SKILLS & EXPERTISE

- **Programming Languages:** Python, C, C++, JavaScript, HTML & CSS
- **Mechanical Design:** SolidWorks, SolidEdge, AutoCAD, MATLAB
- **Frameworks & Libraries:**
 - **Web Development:** ReactJS, Next.JS, Tailwind CSS, Flask, Django
 - **Machine Learning and Data Science:** Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, MediaPipe, NumPy, Pandas, Matplotlib, Deep Boltzmann Machine (DBM), Convolutional Neural Networks (CNN)
 - **i18n:** Internationalization with React
- **Software and Tools:** Raspberry Pi, MATLAB, AutoCad
- **Tools & Platforms:** Arduino IDE, Firebase, MongoDB & Mongoose, Node.js, Git & GitHub, TensorBoard, ImageDataGenerator (Keras)
- **Embedded Systems & IoT:** ESP32 programming, Sensor integration, Arduino IDE, Microcontroller interfacing, Real-time machine monitoring, IoT data communication, Firmware development
- **Soft Skills:** Communication, Teamwork, Problem-solving, Analytical Thinking
- **Languages:** Sinhala, English, and Tamil

PROFESSIONAL EXPERIENCE

- **Undergraduate Trainee** July 2025 - Oct 2025
C.W. Mackie PLC - Scan Division, Horana, Sri Lanka

Key Responsibilities and Achievements

- Designed and modeled mechanical components and assemblies using **SolidWorks** for internal machinery and product development.
- Created and revised detailed technical drawings and layouts using **AutoCAD**, adhering to engineering standards.
- Prepared and maintained **Engineering documentation**, including design reports, Bill of Materials (BOM), and part specifications.
- Collaborated with cross-functional teams to improve equipment performance and reduce downtime.

EXTRACURRICULAR ACTIVITIES

- **Public Visibility Head** 2023 - 2024
IEEE Computer Society Uva Wellassa University
Student Branch Chapter
- **Director of Creation Team** 2024 - 2025
Career Guidance Club of Uva Wellassa University
- **Volunteered in,**
 - **IEEE Open Day** 2024
Head of Design Team
 - **ElectroQuesta 1.0** 2024
Organized by IEEE Robotics & Automation Society
Uva Wellassa University Student Branch Chapter

REFERENCES

DR. THUSHARA EKANAYAKE

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